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SCHOOL OF ARCHITECUTURE AND URBAN PLANNING

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DOCKLESS BIKE-SHARING SYSTEM AND ITS RELATIONS WITH PUBLIC SPACE: WHAT DOES MEDIA NEWS TELL US?

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Abstract: *The dockless bike-sharing system is often owned and operated by the private sector, but ineluctably exerts direct influences on the urban public space and draws abundant public attentions. It creates new ways of space appropriation by promoting cycling in an access-based way. At the same time, problems arise pertaining to spatial conflicts between different users and the maintenance of public spaces. Then, what exactly public perceptions and debates on these issues are? And how would they inform the future of urban public space? This article aims to explicate the social discourse of dockless bike-sharing system, and its relations with urban public space based on media news. Singapore, where the dockless bike sharing experienced the first wave of boom and retreat from January 2017 to April 2019, is used as the case study. A total number of 229 news articles in the city's two major news media are used as analysis samples. Automated-mapping and manually coded content analysis are combined to reveal important issues, the relationship within them and debates among stakeholders, following by a discussion on relations between dockless bike-sharing system and key critiques in public space. The result shows that public perceptions on dockless bike-sharing system differ widely and the relations between dockless bike-sharing system and public space is overlooked. It can be inferred that the way a social group engaging with urban public space significantly influences their tendency on how it should be designed and managed. The article concluded that better public-private partnerships are urgent to improve the dockless bike-sharing system and create a less exclusive urban public space.*

Keywords: *Dockless Bike Sharing, Public Space, Media News, Content Analysis*

00 INTRODUCTION

The so-called sharing economy has experienced remarkable growth in recent years. Although there is no solid definitions of the sharing economy, some characteristics can be summarized, like new economic opportunities disrupting existing markets, access over ownership to goods and services, Information and Communication Technology-enabled interactions and motivation of monetary or non-monetary benefits (Chan and Zhang 2020). Peer-to-peer and business-to-customer are two basic typologies of sharing economy. The sharing economy is believed to have potential in increasing the utilization and efficiency on asserts, making goods and services more affordable, creating flexible employment,

building social networks and communities, and contributing to sustainability. However, scholars indicate sharing economy creates unregulated marketplaces and hit related markets (Martin 2016). It increases inequality because of discrimination behaviors and uneven distribution of its benefits (Frenken and Schor 2017). The environmental effects are also complex, as better accessibility and lower prices make people consume more energy (Schor 2014). Without joint possession, users are likely to solely enjoy the benefit, but be unable and not responsible to maintain the common goods (Chan and Zhang 2020).

The dockless bike-sharing systems (BSS) are for-profit and business-to-consumer platforms run by private operators in the sharing paradigm (McLaren and Agyeman 2015; Schor 2014), but it impacts public realm significantly and draws abundant public attentions. Enabled by smartphone apps and GPS technology, BSS creates new a low-carbon possibility to solve the last-mile travel problem. As a typical disruptive innovation, BSS operators grow rapidly but fail to improve their services and build an effective business model (Si et al. 2020). BSS also brings problems including theft, vandalism, irregular parking and early dumping, which inhibit the sustainability it promises. Chan and Zhang (2018) classified BSS as “urban sharing”, the largest one of the three vectors connecting space and sharing. Urban sharing describes sharing practices widely distributed at the scale of a city, and creating novel issues and social challenges. The direct sharing of bikes has to presume the indirect sharing of urban public spaces, in which the use and maintenance of public spaces have complex conflicts between stakeholders involved in public spaces. Many studies recommend better collaborative regime, integrated social actors, and public-private partnerships as solutions to problems BSS bring (Ma et al. 2018; Shi et al. 2018; Sun 2018). Hence, diverse aspects of values from stakeholders and social groups should be understood for better policymaking (L. Zhang et al. 2015). It is identified as a key research question by Frenken and Schor (2017).

However, there are still limited findings on what exactly public perceptions and debates are on BSS and other sectors of sharing economy. Most existing studies of BSS focus on the perspectives of travel behaviors and mode choices in quantitative approaches (Shen, Zhang, and Zhao 2018; Xu et al. 2019; Zhu et al. 2020). Some studies develop frameworks and conduct case studies in analyzing sustainability, business model and social relations of BSS (Shi et al. 2018; Si et al. 2020; J. Zhang et al. 2017). Especially, most existing studies do not consider establishing relations of BSS with urban public spaces. As such, an opportunity to understand the status-quo, the social discourse and stakeholders’ values of BSS is through the media news. News articles raise public attention and reflect public knowledge, or what was going on, at the current stage (Cheng and Edwards 2019; Schmidt, Ivanova, and Schäfer 2013). News articles also serves as a platform expressing information and viewpoints from different stakeholders. The news media discourse could offer insights into sharing economy and inform researchers and practitioners in identifying directions (Cheng and Edwards 2019). A few studies related to social discourses and tensions in sharing economy paradigm focus on sharing hospitality platforms (Cheng and Edwards 2019; Laurell and Sandström 2017). One research addresses users’ observation of BSS using social media data (Yin, Qian, and Shen 2019).

In this article, we aim to explicate the issues and stakeholders’ debates of BSS and its relations with public spaces based on news streams. To analyze media news, we combined an automated-mapping approach and a manually coding approach in content analysis. Singapore, where the BSS experienced the first wave of boom and retreat from January 2017 to April 2019, is used as the case study. From January to March 2017, three major companies, Ofo, Obike and Mobike launched in Singapore, introducing a few thousand bikes. The number of shared bikes grew to over 100,000 in the following year, operated by them and a few smaller companies. In mid-2018, the government released a new regulation with a license scheme, which

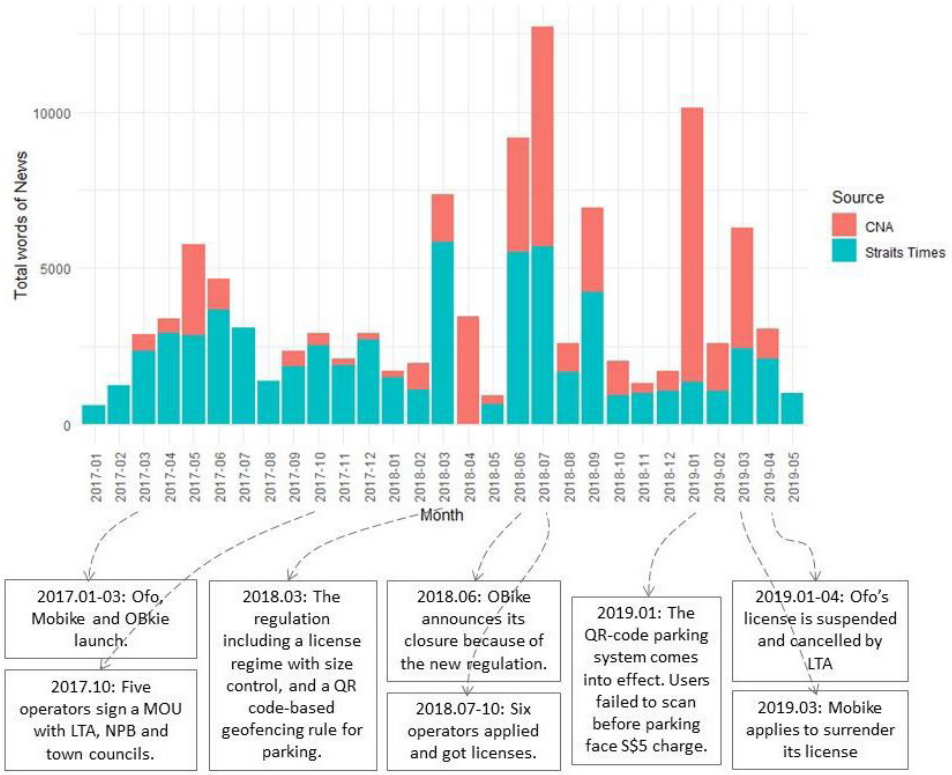


Fig 1.
Total length of news articles in CNA by month and key milestones of BSS in Singapore
Source: Authors

would control fleet numbers of the operators and applying geofencing parking rules to restrict parking at designated places. According to licenses for six operators announced in October, only 40,500 bikes were legal in Singapore. In July 2018, February 2019 and March 2019, Obike, Ofo and Mobike quitted the market of Singapore. Clusters of news articles show the timeline of events in the journey of BSS (Fig 1).

01 METHODS

1.1 Data collection

Online news articles on Singapore's two major media, Channel News Asia (CNA) and Straits Times, are used for this analysis. The news collection was conducted in google advanced search on February 2020. "bike", "Singapore", "sharing" OR "shared" were used as the keywords. Websites were restricted in the official website of CNA and Straits Times, and date period was restricted from Jan 2017 to Jul 2019. URLs of 300 articles in Straits Times and 180 articles in CNA were collected. Through rough skimming the title and the texts, 142 articles in Strait Times and 109 articles on CNA were excluded because their content was not directly related to BSS (e.g. scooter sharing, general cycling collision, electric bikes, cases in other cities rather than Singapore). Then a python script was developed to scrap the full content of these articles. As a result, 158 news articles from Straits Times and 71 articles from CNA were retrieved.

1.2 An automated content analysis by Leximencer

Some studies have found a certain level of efficiency and effectiveness in automated content analysis (ACA). ACA deploys a series of statistical algorithms to uncover hidden patterns and thematic compositions

in a body text. Scholars also applied Leximancer, a widely used ACA software in analyzing literature and news articles about sharing economy in tourism and hospitality (Cheng 2016; Cheng and Edwards 2019). Leximancer have been employed in analyzing academic literature, newspaper article and social media data on multiple topics (Chu, Park, and Kremer 2020; Randhawa, Wilden, and Hohberger 2016).

Leximancer (LexiPortal edition 5) is used to perform ACA in this study. Based on the co-occurrence of the defined concepts, the platform explores the frequencies and relationships of these concepts within defined categories. Through an emergent iterative and unsupervised process, it can quantify the texts and predict what happens in the system (Smith and Humphreys 2006). Firstly, the software automatically learns the preliminary concepts from the texts based on frequencies by excluding some weak semantic terms, says, stop words. We maximize the concept limitation to 100, to consider more possible concepts. Secondly, we organize auto-generated concepts by combining synonyms and adding some concepts not defined but we think they are important. For example, names of operators are combined under the “operator” concept. Finally, the software quantifies the relationship between concepts based on their co-occurrence within context blocks, and represents these relationships in a concept map, which is called the topical map. The size reflects the frequency of concepts, while the same color and proximity of concepts present a strong semantic relationship (Angus, Rintel, and Wiles 2013).

Besides the concept map, a dashboard quadrant is generated, as a method to compare the differences between categories. Based on the frequencies of concepts, the quadrant shows two variables, strength (possibility of a concept falling in a category) and relative frequency (a concept’s frequency compared to other concepts in each category). We set up three categories, government, operators and users, which are three major stakeholders. For better visualization, the quadrant analysis is re-conducted in R using the data retrieved from the Leximancer. Thirty most-frequent concepts of each category are selected for visualizing the quadrant.

1.3 A manually coded content analysis

Based on the ACA above, manually content analysis is conducted using MaxQDA. All 71 news articles from CNA were used as the samples. We used a coding framework including two themes, 1) observation or measures, which are developed in an inductive approach during the coding process; 2) perception of stakeholders, interviews of government departments, operators, users and residents (including specialists), which are four categories. For coded segments of each stakeholder category, ten most frequent concepts are recorded. We also recorded content directly or potentially related to public space for discussion. The manually coding plays as a supplementary and verification for the ACA.

02 THE SOCIAL DISCOURSE OF BSS IN SINGAPORE’S MEDIA NEWS

2.1 Focusing areas of BSS on news media in Singapore

As Leximencer’s topical map (Fig 2) shows, there are four major areas within the news articles, according to the concepts mapping, including 1) nature and economy; 2) governance; 3) shared bike usage and 4)

public space and transportation. Many news articles discuss BSS on the financial, economic and business scopes, in which demand and supply of the market are key focuses. The BSS caught attention once it launched in Singapore, as its technology-enabled features brought lots of potentials. Different from other sharing economy sectors, they provided not only services but also bikes, which was a huge investment. Scholars indicated that the market was oversupply with many operators, resulting in a low utilization of each bike. Moreover, operators competed with each other and attract users by low prices, so they faced difficulties in promoting technologies and services and responding to the government's requirements (e.g. solving the problem of errant parking). This business model was widely regarded to be unprofitable and unsustainable.

The second focus area is the management and legislation from the government departments, mainly Land Transportation Authority (LTA) and town councils. It closely relates to indiscriminate parking, which is the problem catching most attention in news articles. The governance and measures focused mainly on the parking problem. The chronological measures included the following ones: Firstly, LTA fined and impounded errant-parking bikes if operators did not remove them within a half day after notice. Secondly, operators, public enterprises and governments reached an agreement, in which operators should adopt geo-fencing technology and LTA would build designated parking zones to regulate parking behavior. Thirdly, a function was developed in OneService app (the most commonly used app for public service in Singapore) for residents to report inappropriate bike-using behaviors. Finally, a new legislation and license scheme launched, composing a fleet number control, a designate parking zones with QR codes system, and penalties towards users. Following the license scheme, the three biggest operators, Obike, Ofo and Mobike quitted the Singapore market in a half year because of their financial difficulties and ineffective management of operating problems.

News articles revealed users' experiences on the shared bikes, which are closely and directly connected to concept "users" in the map. App is not only the major interface, but also the context where BSS emerged. It is a similar business to regular bike rental and hit them. Some users were blamed for the mistake of inappropriate behavior, mainly misused behavior (e.g. break it, privatize it) and errant parking (e.g. park in

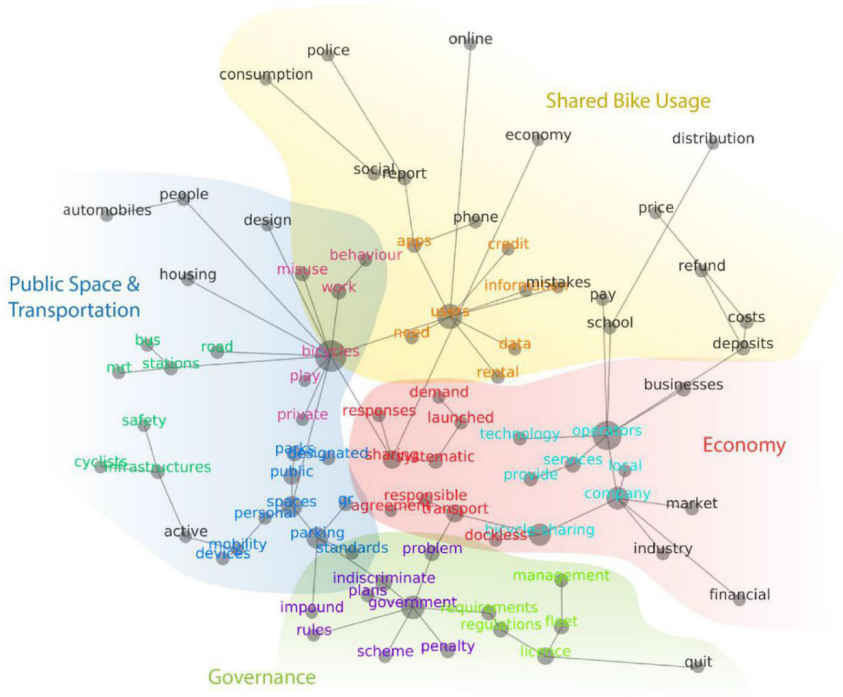


Fig 2.
Conceptual map of all news articles about BSS
Source: Authors

lost space or block sidewalk). In the early stage, most operators used credit systems to guide users' behavior. From the view of government and public, this measure seems not getting an ideal effect. The protection of travel behavior data and users personal information was a public concern. A part of the agreement between operators and LTA was data sharing, which was indicated to contribute to future plans. When the operators withdrew from the market, the spotlight changed to the refund of users' deposits. Many users faced difficulties in getting the deposits back due to the financial problem of operators.

As BSS affect urban public space significantly, the discourse about public space and transportation is another focus area. Singapore has advanced public transport network and hopes to build a car-lite city, but its road network was planned on auto-oriented concepts in the modern age. With limited cycling tradition and infrastructures, Singapore is also trying to bring cycling into its transport plans and to promote its low cycling rate. The dense areas of shared bikes are MRT stations and bus stops, where users transfer to public transit for long-distance travel. The smooth transfer and enough parking zone are the core issue in infrastructural design. Parking is the obvious problem reflecting the impact of BSS on public space. To governments and many residents, the so-called indiscriminate (or errant) parking of shared bikes broke organized and clean public spaces in Singapore.

Besides concept clusters and relations, a deeper exploration of what issues or topics are discussed in news media is necessary. We summarize the coding segments of the topics, including measures government and operators taken, and observatio•n of different prospects of BSS (Table 1). The first three topics are as follows. Firstly, withdraw of three major operators drew most attention in the CNA news (Obike quitted in Jul 2018, Ofo was suspended in Feb 2019, and Mobike quitted in Mar 2019), because there were many related problems, like fail to refund users with their deposits, and lessons learned from the first-wave

Table 1.
Focus issues of BSS in news articles from CNA; Source: Authors

Issues	Total words of News
Operators quit, shut down or license cancelled	11406
Parking box and geo-fencing under license scheme	5623
Market analysis: demand, supply, utilize and profit	5563
License scheme controlling fleet number	3913
Indiscriminate parking	2532
Vandalism	1445
Operators apply credit system and deposit	1054
Agreement by LTA, National Park Board, town councils and operators	1005
Operators collaborate with public enterprise	888
Cycling in Singapore	746
Infrastructure demands and plans	732
Operators get into the market	490
Cyclists' travel behaviors	449
Operators applying more technologies on shared bikes	337
Data sharing	322
Government impound and fine errant-parked bikes	234
An app for residents report indiscriminate parking	161
Volunteers help move errant-parked bikes	106
Operators remove and organize shared bikes	101
Accident and insurance	56

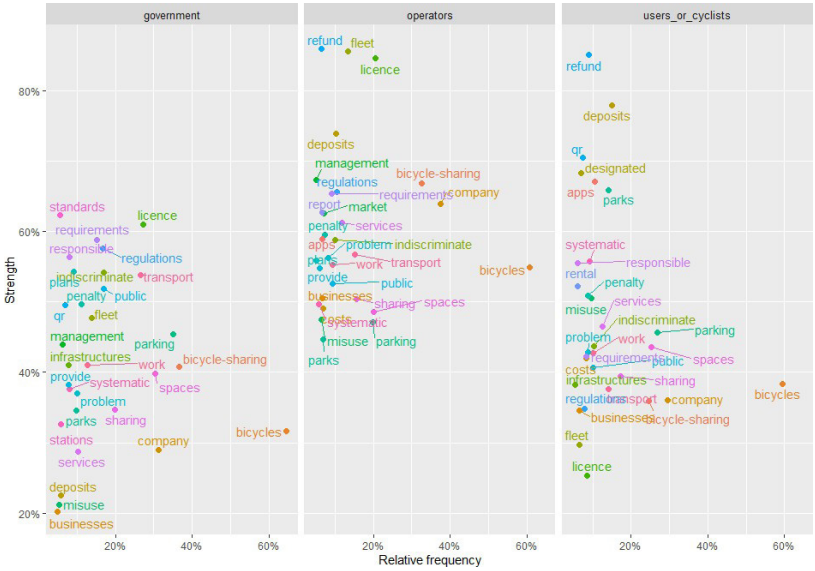


Fig 3.
Comparison between concepts related to each stakeholder in all news articles
Source: Authors

BSS. Secondly, the new license scheme, including the designated parking regulation and the fleet number control, was the other major topic. The third one was about market analysis. CNA invited many scholars to talk about the market and economy of BSS. After these three most frequent topics, two major problems, indiscriminate parking and bicycle misuse by vandals, were identified by CNA. Following was a series of measures, including operators' credit systems, agreement by governments and operators, etc. There were also reports regarding cycling, users' behavior and infrastructures. The manually coded results correspond to the automated map above.

2.2 Comparison between stakeholders on their focuses on BSS

The comparison between stakeholders is also conducted on ACA approach and manually coded approach. The quadrant chart (Fig 3) visualizes thirty most frequent concepts related to each stakeholder, including governments, operators and users or cyclists. The strength measures the unique level of the concept in the category compared to other categories, while the relative frequency shows a concept's frequency compared to other concepts within a category. Different stakeholders have different emphases towards BSS. For example, license and fleet are unique among operators, because they are sensitives about their market share, but the license scheme grand a smaller fleet number than before. Users are more unique in their direct experiences, including refund, deposit, QR, apps, designate (parking zone), which is unsurprising. For the government, standards, requirements, license and responsible remain the unique concepts.

Table 2 compares the frequency of the ten most frequent concepts in each stakeholder's coded segments, 25 concepts in total. The number shows the frequency of concept in each category, divided by total words of segments in this category. This is an accurate way compared with the approach above. Similarly, a more significant difference between the three stakeholder categories is found in this analysis. Users mention more concepts related to their experiences in travel and cycling, like bus, MRT, QR, service and price. Residents, most of them are actually interviewed scholars, focus on business, market, and their understandings on the concept of sharing. Operators mention users a lot, indicating users are their center, and they are willing to improve user's experience, for more market share. The government shows the most comprehensive picture, with mentioning most concepts emphasized by other three stakeholders. It corresponds to the above analysis in which the government has the lowest overall unique level.

Concepts	Government	Operators	Users	Residents
App	0.10%	0.24%	1.17%	0.13%
Bike	4.47%	4.03%	4.00%	5.78%
Bus	0.19%	0.31%	0.75%	0.13%
Business	0.19%	0.37%	0.00%	0.79%
Company	0.68%	0.92%	0.50%	1.97%
Ensure	0.68%	0.24%	0.17%	0.09%
Indiscriminate	0.39%	0.43%	0.00%	0.83%
Fee	0.49%	0.67%	0.08%	0.13%
Fleet	0.87%	0.86%	0.00%	0.22%
License	1.75%	0.67%	0.00%	0.04%
LTA	3.79%	1.10%	0.08%	0.31%
Market	0.29%	0.24%	0.08%	0.70%
MRT	0.29%	0.06%	0.58%	0.04%
Operate	0.68%	0.55%	0.17%	0.66%
Operator	1.85%	1.22%	0.17%	0.88%
Parking	1.65%	1.16%	1.83%	1.18%
Public	1.26%	0.24%	0.08%	0.31%
QR	0.49%	0.55%	0.92%	0.00%
Service	0.19%	0.06%	0.58%	0.18%
Sharing	1.85%	1.10%	1.33%	3.06%
Size	0.68%	0.43%	0.00%	0.18%
Transport	1.26%	0.67%	0.67%	0.79%
Use	0.97%	0.43%	2.33%	0.74%
User	0.78%	2.81%	1.67%	0.88%
Price	0.00%	0.12%	0.75%	0.13%

Table 2.
Comparison between stakeholders' focused concepts in coded segments of CNA news

03 THE RELATIONS OF BSS AND PUBLIC SPACE

In this part, we discuss the relations of BSS and public space based on information in media news. Firstly, BSS involves in urban public space and have significant impacts, while public space facilitates the operation of BSS. Secondly, the occurrence of BSS help perceive original problems of public space, and reveal differences of defining public space and proper ways to use it between social groups. The use of shared 26 bike, a semi-public facility operated by private companies, reflect public attitude towards public goods. The discussion is developed following a framework of key critiques about urban public space.

3.1 The key critiques on contemporary urban public space

Carmona (2010) summarized the critiques of public space by classifying them into two broad camps, under-management and over-management. Several sub-typologies were revealed in each camp. However, he also pointed out that under management and over management were two sides of the same coin with each contributing to the other. In other words, there were lots of overlaps in the causes of these public-space typologies. Thus, we try to apply another way of organization, extracting key critiques and their causes, which is for establishing a discussion framework connecting public space and BSS.

The key critiques fall into eight aspects.

1) Poor design causes physical barriers and obstacles, leading to exclusion for some specific social groups who are sensitive to the environment, such as seniors, kids and people with disabilities. Some design strategies are unfriendly to certain groups and play as restrictions to them subjectively or objectively. For example, suburb public facilities and amenities become fortresses surrounded by moats of car parking or high blank walls, which is low accessible to pedestrians.

2) Poor management and after-care because who should be responsible for management are rarely clear. After minor signs of decay appeared, space would be brought to a rapid spiral of decline with less and less usage, just as a broken window. Then it is neglected or lost, missing the opportunities to bring together disparate activities and connections.

3) On contrast, some public spaces are over-managed with high-level cleanness and security in fear of strangers, crime and decay. More surveillance and control on activities are brought, including hard (active) and soft (passive) ways. For instance, using symbols, place atmosphere designs and visual cues as restrictions to undesirable people (e.g. those cannot afford) and activities.

4) Spatial segregation happens in urban space because of suspicion, tension and conflict between different social groups. Some distinct spatial types form to serve specific users. Gated communities, evening-economy streets and gay villages are examples.

5) Besides over-management, another consequence of under-management is turning truly public space to private and semi-private domains. With the shrinking public realm, some public spaces are abandoned to market forces for consumption. In turn, privatization and commodification aggravate exclusion and over-management.

6) Under the modern movement, land-use separation in zoning and dominant of automobiles slice, degrade and fragment public space. In this case, private automobiles are invaders to public space. Public spaces are simply flown past by car and lose social functions. After being ignored for a long time, external public space even disappeared and turn to indoor enclaves (consumption space).

7) A further result is reluctance of parents to let their children use public space, like playing on street or walking to school, because of the safety concerns. This kind of lifestyle and experience push teenagers farther from public space revival. Public culture declines along time.

3.2 Discussion of relations between BSS and public space

The physical barrier and exclusive design are mainly reflected in the lack of cycling infrastructures. Although in transport plans of Singapore, cycling has been figured out as an important last-mile transport, the infrastructure is insufficient seriously (Land Transport Authority Singapore 2019). There are limited bike lanes or bike paths within the city. Cycling on sidewalks in less than 10km/h is legal (Land Transport Authority Singapore 2020), but narrow sidewalks cause conflicts and inconvenience, and often push cyclist to the vehicle lanes. The unfriendly street environment is a major barrier to cycling. In this way, the development of BSS has limited support from public space.

The operation of BSS showed signs of poor management in operators' business model, public space maintenance and users' right protection. These signs included fail of refunding deposits and clearing bikes while quitting, some misuse behavior, and so-called indiscriminate parking. The major reason was operators'

own business paradox - the limited cycling culture, oversupply, low prices for competition, and low utilization of each bike. As the outcome, they were financially unsustainable and not able to meet the government's standard of managing fleets (Lim 2018). The poor management threatened the bike-sharing industries and left negative impressions to residents. In addition, those improper behaviors on shared bike is comparable to similar behaviors on other public facilities. They could happen in all societies and requires similar managements. Furthermore, the spaces people breaking or throwing bikes are usually under- managed public spaces (in some western countries, usually in depressed neighborhoods, BBC News 2018). Misuse of shared bikes identifies them unintentionally. Those lost or neglected spaces themselves need our attention.

In order to avoid BSS and public space from a spiral of decline, LTA employed series of measures facing indiscriminate parking. The measures included a strict remove (within a half day), fine (S$500) and impound rule to operators, a new app function for residents reporting indiscriminately parking, and finally, a license scheme with fleet number control, designate parking spaces, and penalty to users (S$5 fine for each time parking errantly and one-year ban for three times). These measures can be classified as hard control. There were debates about whether are over-managed. For users, some expressed that the punishment under

Table 3.
Relations of key critiques in public space and BSS

Key critiques in public space	Implication in BSS
Physical barriers and re-strictions form due to poor or unfriendly design	Lack of infrastructure and unfriendly street environment are major barriers
Public space becomes de-cline, neglect or lost because of poor management	Private operators without regulation show signs of poor management
	Vandal behaviors of shared bike identify under-managed public spaces
Fear of strangers and crime lead to over management on cleanness and security and exclusion of undesirable people or activities	The strict measures and regulations harm the mobility of BSS
	There are possibilities to guide fetching behaviors and balance mobility and utilization
	The long-time obsession of cleanliness and discipline is an internal dynamic for over-management, which somewhat conflicts with promoting cycling and innovation
Conflicts between social groups cause spatial segregation and distinct spatial types	Different groups have significantly different perceptions on the desirable way to use public space
	How to distribute the street space to create an inclusive environment for multiple transport modes
	The shrinking of public realm, commodification and privatization of public space aggravate over-management, exclusion and restriction
	The process of management transition is different to western countries because of a stronger state capacity
	A better public-private partnership to be explored to reach a balance point between under and over management
Separated land uses and dominant private automobiles fragment and even eliminate outdoor public space. Private automobile is the invader.	The physical planning in Singapore shape residents' travel habits and attitudes towards BSS: who is the invader, shared bikes or private automobiles?
	To change residents' values, walking-and-cycling-centered planning strategies should be employed in the future
Public space loses social func-tions and public culture declines	Singapore's public culture can still be promoted on street-life revival, where BSS can play a role
	Cultural and education programs will help encourage cycling and build social networks among bike-sharing users

the license scheme was harsh (Ho 2018). Operators expressed difficulties in removing bikes within a half day (Abdullah 2017). BSS contributed mostly to the last-mile travel, but following the new rule, parking was allowed only in designated parking boxes. This rule weakened the mobility of BSS. Many cyclists indicated they would stop to use shared bike because the new rule led to less convenience (Wong and Ng 2018). On the contrast, when shared bikes can be parked anywhere, many of them would stay in some neglected space with limited access and usage – it is still about the low share rate of cycling, or insufficient users and cyclists to support a profitable operation. In this case, there are possibilities in guiding and encouraging users to use nearby neglected bikes by applying specific algorithms. For example, if a bike has not been used over a liminal time, it will get a special tag. Who use it can get extra credits although it is not the nearest bike. This kind of algorithm, known as rebalancing dockless bikes, has become a popular research topic(Li et al. 2019; Zhu et al. 2020).

A more internal reason for the over-management (if there is) is the long-time obsession with cleanliness and discipline. When talking about indiscriminately parking, what people care most were not bikes in neglected spaces but those blocking sidewalks and cluttering on green strips. Cyclists argued that bikes were more eyestones than real obstacles for residents and the government (Tan 2018). In Singapore, strict cleanness, order and security in the public space has been societal norms after decades of clean campaigns (McDonald 2018). The country believes that a well-managed environment creates a stronger economy. The societal norm formed an atmosphere of soft control on bike parking behaviors. Nowadays, more conflicts arise between this social norm and some development goals, including a car-lite city promoting cycling, and an innovation culture encouraging ideas. The BSS reveals this conflict. BSS has been identified as an innovative solution by Singapore's transport plans, but the city will experience short-term negative impacts. The country might need more flexibility and less preoccupation on orderliness and discipline in public space, to avoid shackling innovation (Agarwal 2018).

The conflict among social groups and the spatial exclusion make the social environment less friendly to BSS. As mentioned above, the focus points of BSS have large differences among the four identified social groups, residents, users, operators and the government. Furthermore, their perceptions vary about what are the desirable ways of using public space. Cyclists thought BSS was the future because it is greener, while many residents were upset that shared bikes broke the city's orderliness (Wong and Ng 2018). The somewhat unfriendly social environment suppressed bike-sharing usage and even cycling. Additionally, the conflicts among groups lead to a spatial issue, distributing street space for different transport modes. No moving space and insufficient parking space excluded cycling as a transport mode. Encouraging mode shift to active transportation requires street redesign to assign cyclists and pedestrians more space (e.g. bike lanes, crossing markings and signals for cycling). These designate spaces remind cyclists of an inclusive environment.

In the BSS case in Singapore, the transition between public and private realm is different from western countries where public space theories origins. In those countries where local government shrinks and public realm declines, some public spaces become poorly managed and are thrown to the markets by increasing private ownership. Then some over management and exclusion happen. However in Singapore's case, free access to public space was granted to private operators at the beginning. Operators used public space and facilities for profits without rent, which attracted many critiques. The aim of private corporations is making profit, so they have limited willing to maintain public space. When some signs of poor management appeared, the government involved and negotiated with operators, and implemented a license scheme.

Since then, the management became strict with operators and users. This process shows Singapore has stronger state capacity to manage public space.

Although the government plays the key role, a proper public and private relationship helps avoid under or over management. It was discussed in media that whether market consolidation, like having two or three companies, can be good in certain periods (Varakantham 2018). Some also argued that Singapore may go back to a public bike-sharing scheme after the retreat of three big operators (Lin 2019). However, we think between government, operators and social groups, establishing a public-private partnerships is necessary. It is to ensure policies and regulation reaching a balanced point between under and over management. To encourage cycling, government can provide incentives, subsidies and infrastructures. There should also be regulations to protect users' interest (e.g. avoid situations like operators cannot refund deposits). Operators should compete by employing better technologies and improving service.

Singapore's physical planning shapes residents' travel habits and attitudes towards BSS. Physical planning in Singapore has been deeply influenced by modernism principles. The land-use separation, superblocks, overpasses and cul-de-sacs can be seen in most residential towns. It did happen that street networks designed for private automobiles sliced and fragmented public space. The city's planning rarely considered cycling. However, these towns have high density, sheltered walkways and convenient public transport. The city also implemented restrictions on ownership of private cars. Therefore, the private automobiles is not that dominant comparing to most megacities. The first three common modes, public transit, private automobile and walking, share over 90% of travel in Singapore (Land Transport Authority Singapore 2019). Residents have kept existing travel habits for a long time and not seem to have strong willingness to change. Thus, we can read from news that residents regard BSS rather than cars as the invader by residents, especially when BSS are parked irregularly on streets (Peng 2018; Tan 2018). The different definitions on invaders closely relate to the judgement on whether over management exists, and perceptions on public-private relationships. In the future, if the city hopes to change residents' values, different planning and design strategies should be employed, including mixed land-uses, better street networks and more attractions on street facades.

The public culture in Singapore still can be promoted along the process of street-life revival, where BSS can play a role. The city should cultivate the cycling culture through education programs if it hopes to change the cognition of residents, improve the share rate of cycling and create more opportunities to BSS. In the year 2017, operators, town councils and schools had planned collaborations in such education programs, but these programs did not continue in the following years. The above-mentioned public-private partnership should involve in culture and education. More importantly, promoting cycling is about developing a public culture on the street. Residents in Singapore have socialized lifestyles and habits of riding public transport. In most residential towns, there are some active community spaces (e.g. hawker centers), but they are centralized and homogenous. Most neighborhood streets lose their social functions and become simply moving spaces. Reassigning social and recreational activities to streets requires different land-use planning, environmental design and program organization. Revitalizing streets as linear public spaces benefits community bounding. The transportation nodes, including bus stops and bike stations, have large potentials in mixed and social functions. BSS itself can be an opportunity in building social connections, networks and communities. Many studies pointed out that social interaction with strangers is always a major motivation of participating different sharing programs (Chan and Zhang 2020). With the help of cultural and educational programs, the networks built by bike-sharing users can contribute to community bounding and public culture revival.

04 CONCLUSION

In summary, this study explores the social discourse of BSS based on news stream in Singapore. We apply a combination of automated and manually coded content analysis to identify the important issues and their relationships. Following this approach, the study also revealed different perceptions between stakeholders on the BSS and governance responses. This approach could be used for future research on social discourse in BSS and other sectors of sharing economy, because they involve complex stakeholder relationships with conflict opinions. In order to manage a healthy development of BSS, efficient public-private partnerships should be established in diverse forms. It is not only for better policymaking and infrastructural planning, but also for education and activities to promote cycling and public manners.

In both academic and news articles, there is a lack of direct and robust narrative on relations between BSS and public space, in which this study aims to offer a starting point. Firstly, the governance towards BSS still swings between under and over management, making mobility seems to conflict with space quality. Secondly, improving rebalance technologies, applying subsidy incentives and establishing cycling infrastructures will improve management and better facilitate BSS and cycling. Thirdly, BSS could be a catalyst for promoting public culture, encouraging innovation and revitalizing the street life. If we can combine planning, design and programs to reshape residents' perceptions and behaviors, we will achieve a virtuous and sustainable relationship between BSS and public space.

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